

PLAIN STRENGTH OF COMPOSITE LAMINATES

Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3403**

Title: **PLAIN STRENGTH OF COMPOSITE LAMINATES**

Revision: **1.1**

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1 INTRODUCTION

1.1 OBJECT

Some words from the Word-Wide Failure Exercise (1st edition) introduction article: “Current commercial design practices place little or no reliance on the ability to predict the ultimate strength of the structure with any great accuracy. Failure theories are often used in the initial calculations to 'size' a component (i.e. to establish the approximate dimensions, such as panel thickness, width, etc). Beyond that point, experimental tests on coupons or structural elements (such as the notched hot/wet compressive strength tests in aerospace) are used to determine the global design allowables.”

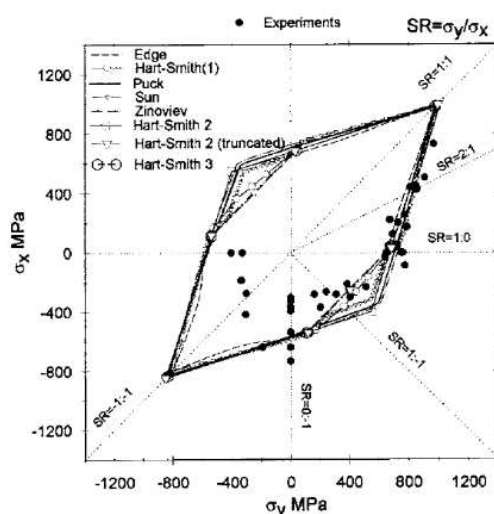


Fig. 6(a). Comparison between the predicted and measured biaxial final failure stresses for (0°/±45°/90°) AS4/3501-6 laminates (Test Case No 6).

Figure 1. Some failure criteria vs tests results in the WWFE study, from Kaddour's paper

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DEFENCE AND SPACE

Conclusions of the WWFE did not close this controversial topic. Even though it was admitted that, when discussing about aerospace standard practices, using typically $0^\circ/\pm 45^\circ/90^\circ$ configurations, *“even the adoption of so called “black aluminium” does not exploit the benefits of the CFRP material to the full, but does simplify the design rules by removing the need to worry about matrix induced failure modes.”*

For practical reasons, for these standard aerospace laminates, one of the failure criteria presented in that study, proposed by L. J. Hart-Smith, the *Maximum Truncated Strain* Failure Criterion, is recommended by the CMH-17. That recommended laminate level failure criterion for carbon fibre composites is a modification of the maximum strain criterion. The same assumptions of no external bending, membrane forces constant along the edges, and a balanced and symmetric laminate, are initially used. The basic lamina failure envelope is a variation of the conventional maximum-strain envelope for tension- and compression-dominated loads. The modification consists of added cut-offs to the preceding maximum-strain failure model, in the tension-compression quadrants.

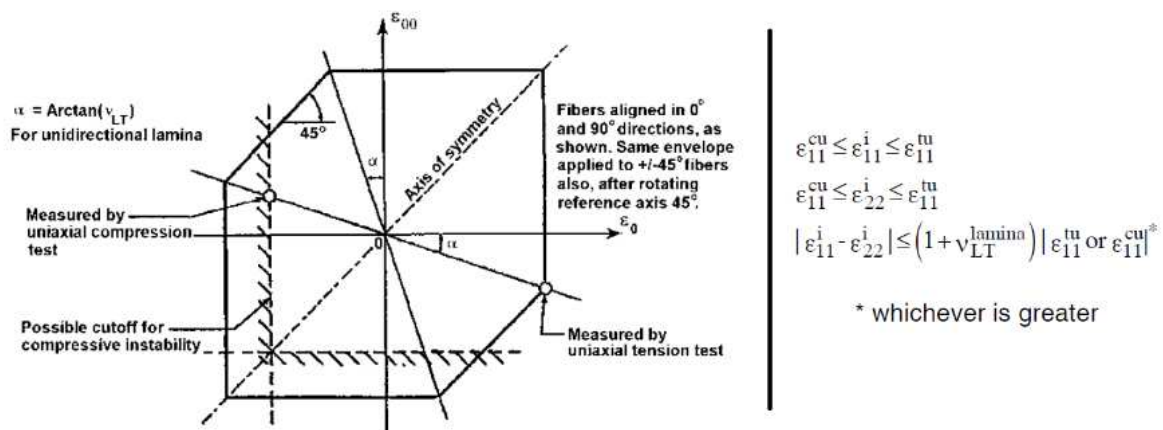


Figure 2. Maximum Truncated Strain failure criterion, from (CMH-17 Vol. 3)

However, it has been *“found that the unidirectional data does not adequately predict the unnotched tension or compression strength of a multi-directional laminate. For this reason, these companies use strength values obtained from multidirectional laminate test data and established as a function of laminate lay-up with the maximum strain failure criteria”* (CMH-17 Vol. 3).

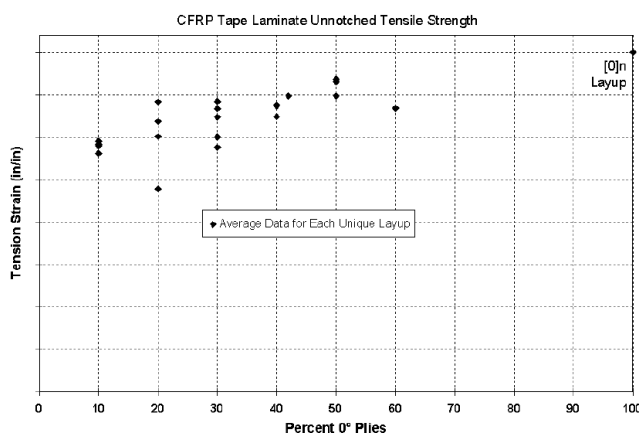


FIGURE 8.6.2.3(a) Example laminate tensile strength as function of lay-up.

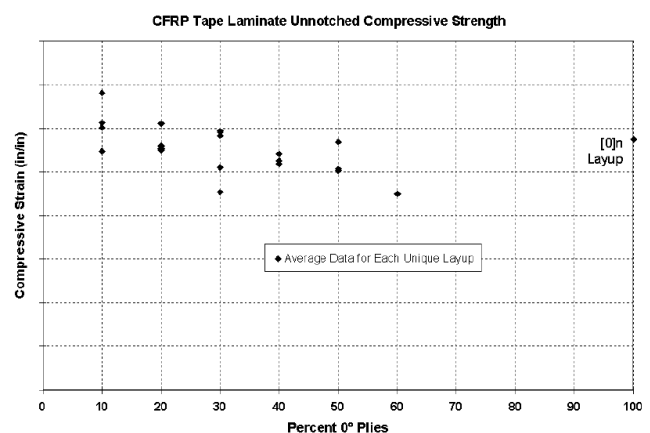


FIGURE 8.6.2.3(b) Example laminate compressive strength as function of lay-up.

Figure 3. Strain at failure in plain tension & compression for different laminates (CMH-17 Vol. 3)

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1.2 SCOPE AND LIMITATIONS

Plain strength of fibrous composite laminates subjected to in-plane and plate loads.

Failures due to through thickness stresses, notched laminates or damage tolerance considerations are out of the scope of this chapter.

2 METHOD DESCRIPTION

2.1 FAILURE CRITERION FOR STANDARD LAMINATES

By standard laminates we refer to balanced and symmetric laminates (quasi-orthotropic), with stacking sequences including at least a 10% of the plies at 0°, 90°, 45° and -45° angles (see Ref. 7, chapter "Good Design Practices and Design 'Rules of Thumb'"). Provided stability issues are elsewhere assessed, for this type of laminates a variant of the Maximum Strain criterion is recommended as failure criterion for plain strength.

Frequently, laminates are fibre-dominated, meaning that there are fibres in sufficient multiple directions such that strains are limited by the presence of the fibres which inhibits matrix cracking. In many practical applications, this typically translates into having fibres in (at least) each of four directions relative to the primary loads: 0°, 90°, and ±45°. Furthermore, plies should not be "clustered" (that is, several plies of the same orientation are not layered together) in order to inhibit matrix macro-cracking. With these assumptions, there is a straightforward translation of a maximum strain ply failure criterion to the laminate level. For these fibre dominated laminates, one approach has been to limit the strain in the transverse direction to the value of the fibre direction limiting strain. This reflects the fact that laminates with fibres in multiple directions restrict strains in all in-plane directions. Presented in the standard notation used in the company, where 1/2 are material ply longitudinal/transverse directions, the abovementioned *Maximum Truncated Strain* Failure Criterion is proposed as design criterion. Laminate fails in any of this conditions is not satisfied:

$$\begin{aligned} -\epsilon_{c-all} < \epsilon_1 < \epsilon_{t-all} & \quad -\epsilon_{c-all} < \epsilon_2 < \epsilon_{t-all} \\ |\epsilon_1 - \epsilon_2| < (1 + \nu_{LT}) \cdot \max(\epsilon_{t-all}, \epsilon_{c-all}) \end{aligned}$$

See discussion about loads/strains and design allowable values ϵ_{t-all} , ϵ_{c-all} , ν_{LT} to use in these formulas in following paragraphs.

2.1.1 Ply strains at applied loads

Classical Laminate Theory permits, in first instance, obtain the mid-plane deformation, i.e. the ϵ_x , ϵ_y , γ_{xy} extensional strains and the κ_x , κ_y , κ_{xy} curvature deformations, of a plate under certain loads (see chapter §3401 "STRESS AND STRAIN IN COMPOSITE LAMINATES"). Calculating the strain for each ply, in the laminate reference axis system, is then straight-forward:

$$(\epsilon_x, \epsilon_y, \gamma_{xy})_{ply} = (\epsilon_x, \epsilon_y, \gamma_{xy})_{mid-plane} \pm (\kappa_x, \kappa_y, \kappa_{xy}) \cdot z_{ply}$$

Expressions for computing fibre strains in 0°, 45°, -45° and 90° directions from the strain expressed in the laminate reference axis system can be found in the cheat-sheet of abovementioned chapter.

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A possible simplification of this failure criterion to laminates submitted to moderately low bending loads (with no significant through thickness stresses) is to apply the expression above to the strain levels at the outer faces:

$$(\epsilon_x, \epsilon_y, \gamma_{xy})_{face} = (\epsilon_x, \epsilon_y, \gamma_{xy})_{mid-plane} \pm (K_x, K_y, K_{xy}) \cdot t / 2$$

2.1.2 Allowable strains of the material

Allowable strains in tension and compression shall be obtained from the test data available from material qualification, for the unidirectional laminate and for multi-angular laminates. Typically at least the QI laminate should have been tested, but validation of the design allowables with test data on soft (10/80/10) and hard/directed (50/40/10) laminates is highly desirable, especially for UD-tape materials in tension.

UD-tape materials

- ✓ Tension allowable strain

Take ϵ_{t-all} as the ultimate strain of the quasi-isotropic (QI) laminate, calculated from the plain strength test of that laminate and its “nominal” Young’s modulus (calculated from the design values for the elastic moduli of the materials and the QI stacking sequence).

It is remarked that this convention may lead to excessively optimistic strength predictions in laminate directions with few fibres aligned with tension load, and thus be invalid. It is expected, however, that laminate will be sized with fibres proportions beyond 25% in directions critically loaded in tension, and this flaw of the method will be finally not relevant for design.

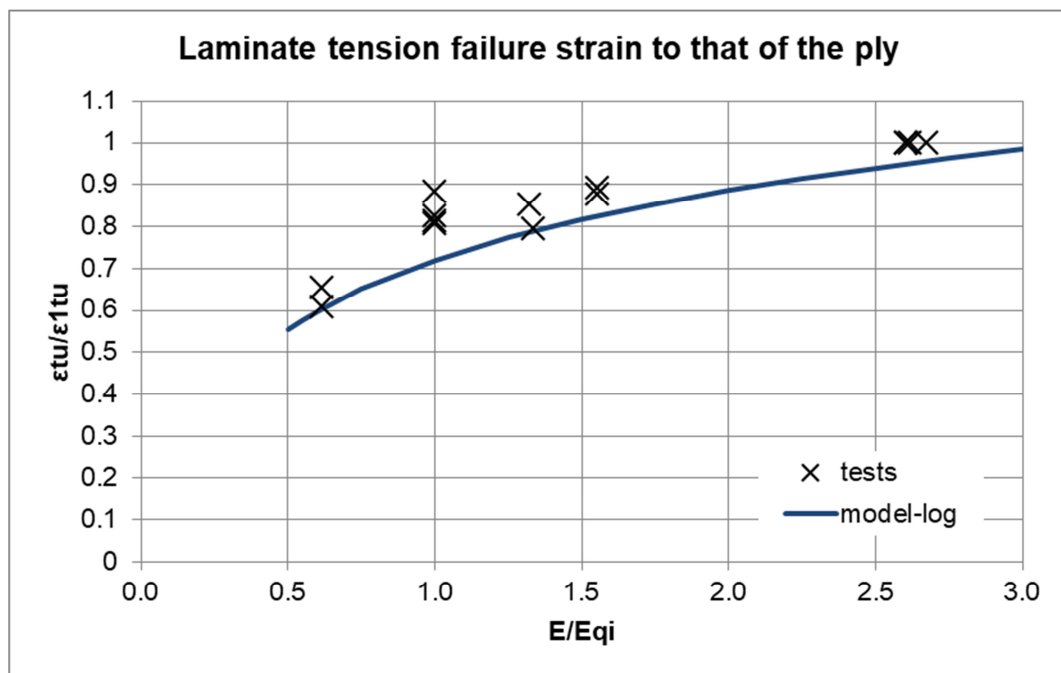


Figure 4. Tension allowable strain of laminates made of UD-tape CFRP materials (tests data points from Hexcel AS4/8552, IMA/M21E, AS7/M21X and Cytec HTS/977-2 coupons)

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(*) It is under consideration (pending discussion, agreement and validation) an approximation of this magnitude capable to account the tension strength variation within the design space (see previous figures, representing ultimate tension strains of laminates, from CMH-17 and in-house tests data).

✓ Compression allowable strain

Take $\epsilon_{c-all} = \epsilon_{1c}$ from the compression strength test data of the unidirectional laminate, in warp direction, as the allowable compression strain of the material independent of the laminate stacking.

This is expected to be conservative (see trend of second graph in Figure 3, confirmed with internal Airbus tests data) and not conditioning the part sizing (failure modes associated to buckling instabilities being generally more critical).

✓ Poisson's ratio

Take the Poisson's ratio of the UD-tape ply material ν_{12} as valid design allowable in the formula ($\nu_{LT} = \nu_{12}$).

Bi-axial fabrics

✓ Tension allowable strain

Take $\epsilon_{t-all} = \min(\epsilon_{1t}, \epsilon_{2t})$ from tests data of unidirectional laminates (tension strength in warp and weft directions) as the allowable tension strain of the material independent of the laminate stacking.

The variation of the tension allowable strain with the laminate directionality observed in UD-tape materials has not been observed in internal Airbus tests, with laminates made of bi-axial fabrics within a range from (QI) 25% to (UD) 50% plies/fibre content oriented in load direction.

✓ Compression allowable strain

Take $\epsilon_{c-all} = \min(\epsilon_{1c}, \epsilon_{2c})$ from tests data of unidirectional laminates (compression strength in warp and weft directions) as the allowable compression strain of the material independent of the laminate stacking

✓ Poisson's ratio

For bi-axial fabric materials, instead of the Poisson's ratio of the material ν_{12} , use directly a value of 0.3 (i.e. treat the fabric ply as a cross-ply 0/90 sublaminates made of a fictitious UD-tape material with the same allowable strains).

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2.2 FAILURE CRITERION FOR NON-STANDARD LAMINATES

In case the laminate does not follow the standard design rules, i.e. is not symmetric, balanced, does not include sufficient plies in 0° , 90° or $\pm 45^\circ$, there are significant through-thickness stresses, etc... a progressive failure analysis methodology is preferred to the (modified) maximum strain criterion described above.

Nevertheless, a preferred point stress failure criterion for evaluating damage on the plies and damage and degradation strategies is still not decided. This document shall be updated as soon as a design practical approach is agreed.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
E	MPa	Young's modulus of a laminate in a direction of interest
E_{qi}	MPa	Young's modulus of the quasi-isotropic laminate
ϵ_{1tu}	-	Ultimate/failure tension strain of a FRP material in 0° /warp direction
ϵ_{1t}	-	Allowable tension strain of a FRP material in 0° /warp direction
ϵ_{1c}	-	Allowable compression strain of a FRP material in 0° /warp direction
ϵ_{2t}	-	Allowable tension strain of a FRP material in 90° /weft direction
ϵ_{2c}	-	Allowable compression strain of a FRP material in 90° /weft direction
ϵ_{tu}	-	Ultimate/failure tension strain of a laminate
ϵ_{t-all}	-	Allowable tension strain of a laminate
ϵ_{c-all}	-	Allowable compression strain of a laminate
ϵ_x, ϵ_y	-	Extensional strains in X and Y directions
FRP	-	Fibre Reinforced Polymer
γ_{xy}	-	Shear engineering strain in X-Y plane
K_x, K_y, K_{xy}	mm^{-1}	Curvature strains in a laminate
ν_{12}	-	Poisson's ratio of a FRP material
QI	-	Quasi-Isotropic laminate, symmetric, balanced, and with the same number of plies at 0° , 90° and $\pm 45^\circ$. It is typically built using a sequence of sub-laminates (45/0/-45/90) and applying symmetry, for example ((45/0/-45/90)3)S
t	mm	Thickness of a laminate
WWFE	-	(First) World Wide Failure Exercise
z	mm	Distance from the laminate mid-plane to a ply

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[M. J. Hinton et al]	2004	The world-wide failure exercise: Its origin, concept and content
Ref. 2	[P. D. Soden et al]	2003	Recommendations for designers and researchers resulting from the world-wide failure exercise
Ref. 3	[SAE] CMH-17	2012	Composite Materials Handbook (Volume 3)
Ref. 4	[L. J. Hart-Smith]	1997	PREDICTIONS OF THE ORIGINAL AND TRUNCATED MAXIMUM-STRAIN FAILURE MODELS FOR CERTAIN FIBROUS COMPOSITE LAMINATES
Ref. 5	[A. S. Kaddour]	2004	Predictive capabilities of nineteen failure theories and design methodologies for polymer composite laminates. Part B: Comparison with experiments
Ref. 6	[A.Puck,H.Schürmann]	2001	Failure analysis of FRP laminates by means of physically based phenomenological models
Ref. 7	[C. Kassapoglou]	1 st Ed.	DESIGN AND ANALYSIS OF COMPOSITE STRUCTURES

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
ANCOLAB (Airbus DS)	4.1	Classical laminates analysis

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 13/11/2019	G. Baños	First issue	All pages
1.1 14/01/2020	G. Baños	<i>Maximum Truncated Strain</i> failure criterion selected as preferred design criterion for conventional laminates	Pages 3, 5

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

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APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEASP-TL3)	G. Baños 29/12/2019	F. Glock 14/01/2020

Table 3. *Approval signatures table for last revision of this chapter*